TABLE 5.1 Fourier Transform Properties

Operation	Time Function	Fourier Transform
Linearity	$af_1(t) + bf_2(t)$	$aF_1(\omega) + bF_2(\omega)$
Time shift	$f(t - t_0)$	$F(\omega)e^{-j\omega t_0}$
Time scaling	$f(at)$	$\frac{1}{\lvert a\rvert} F\left(\frac{\omega}{a}\right)$
Time transformation	$f(at - t_0)$	$\frac{1}{\lvert a\rvert} F\left(\frac{\omega}{a}\right)e^{-j\omega t_0/a}$
Duality	$F(t)$	$2\pi f(-\omega)$
Frequency shift	$f(t)e^{j\omega_0 t}$	$F(\omega-\omega_0)$
Convolution	$f_1(t)*f_2(t)$	$F_1(\omega)F_2(\omega)$
	$f_1(t)f_2(t)$	$\frac{1}{2\pi}F_1(\omega)*F_2(\omega)$
Differentiation	$\frac{d^n[f(t)]}{dt^n}$	$(j\omega)^n F(\omega)$
	$(-jt)^n f(t)$	$\frac{d^n[F(\omega)]}{d\omega^n}$
Integration	$\int_{-\infty}^{t} f(\tau)d\tau$	$\frac{1}{j\omega}F(\omega) + \pi F(0)\delta(\omega)$

TABLE 5.2 Fourier Transform Pairs

Time Domain Signal	Fourier Transform
$f(t)$	$\int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$
$\frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}d\omega$	$F(\omega)$
$\delta(t)$	1
$A\delta(t - t_0)$	$Ae^{-j\omega t_0}$
$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
1	$2\pi\delta(\omega)$
K	$2\pi K\delta(\omega)$
$\text{sgn}(t)$	$\frac{2}{j\omega}$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin \omega_0 t$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
$\text{rect}(t/T)$	$T\,\text{sinc}\,(\omega T/2)$
$\cos(\omega_0 t)u(t)$	$\frac{\pi}{2}[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] + \frac{j\omega}{\omega_0^2 - \omega^2}$
$\sin(\omega_0 t)u(t)$	$\frac{\pi}{2j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] + \frac{\omega_0}{\omega_0^2 - \omega^2}$
$\text{rect}(t/T)\cos(\omega_0 t)$	$\frac{T}{2}\left[\text{sinc}\left(\frac{(\omega - \omega_0)T}{2}\right) + \text{sinc}\left(\frac{(\omega + \omega_0)T}{2}\right)\right]$
$\frac{\beta}{\pi}\text{sinc}\,(\beta t)$	$\text{rect}(\omega/2\beta)$
$\text{tri}(t/T)$	$T\,\text{sinc}^2(T\,\omega/2)$
$\text{sinc}^2(Tt/2)$	$\frac{2\pi}{T}\text{tri}(\omega/T)$
$e^{-at}u(t), \text{Re}\{a\} > 0$	$\frac{1}{a + j\omega}$
$te^{-at}u(t), \text{Re}\{a\} > 0$	$\left(\frac{1}{a + j\omega}\right)^2$
$t^{n-1}e^{-at}u(t), \text{Re}\{a\} > 0$	$\frac{(n - 1)!}{(a + j\omega)^n}$
$e^{-a\lvert t\rvert}, \text{Re}\{a\} > 0$	$\frac{2a}{a^2 + \omega^2}$
$\sum_{n=-\infty}^{\infty} g(t - nT_0)$	$\sum_{n=-\infty}^{\infty} \omega_0 G(n\omega_0)\delta(\omega - n\omega_0), \omega_0 = \frac{2\pi}{T_0}$
$\sum_{n=-\infty}^{\infty} g(t - nT_0) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}$	$H(\omega) = \lvert H(\omega)\rvert\angle\phi(\omega) = \frac{\lvert V_2(\omega)\rvert\angle V_2}{\lvert V_1(\omega)\rvert\angle V_1} = \frac{\lvert V_2(\omega)\rvert}{\lvert V_1(\omega)\rvert}\angle V_2 - \angle V_1,$
$\delta_T(t)$	$\sum_{k=-\infty}^{\infty} \omega_0\delta(\omega - k\omega_0)$

TABLE 7.2 Laplace Transforms

f(t), t ≥ 0	F(s)	ROC
1. $\delta(t)$	1	All s
2. $u(t)$	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
3. t	$\frac{1}{s^2}$	$\mathrm{Re}(s) > 0$
4. t^n	$\frac{n!}{s^{n+1}}$	$\mathrm{Re}(s) > 0$
5. e^{-at}	$\frac{1}{s+a}$	$\mathrm{Re}(s) > -a$
6. te^{-at}	$\frac{1}{(s+a)^2}$	$\mathrm{Re}(s) > -a$
7. $t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$	$\mathrm{Re}(s) > -a$
8. $\sin bt$	$\frac{b}{s^2+b^2}$	$\mathrm{Re}(s) > 0$
9. $\cos bt$	$\frac{s}{s^2+b^2}$	$\mathrm{Re}(s) > 0$
10. $e^{-at}\sin bt$	$\frac{b}{(s+a)^2+b^2}$	$\mathrm{Re}(s) > -a$
11. $e^{-at}\cos bt$	$\frac{s+a}{(s+a)^2+b^2}$	$\mathrm{Re}(s) > -a$
12. $t\sin bt$	$\frac{2bs}{(s^2+b^2)^2}$	$\mathrm{Re}(s) > 0$
13. $t\cos bt$	$\frac{s^2-b^2}{(s^2+b^2)^2}$	$\mathrm{Re}(s) > 0$

TABLE 7.3 Laplace Transform Properties

Name	Property
1. Linearity, (7.10)	$\mathcal{L}[a_1f_1(t) + a_2f_2(t)] = a_1F_1(s) + a_2F_2(s)$
2. Derivative, (7.15)	$\mathcal{L}\left[\frac{df(t)}{dt}\right] = sF(s) - f(0^+)$
3. nth-order derivative, (7.29)	$\mathcal{L}\left[\frac{d^nf(t)}{dt^n}\right] = s^nF(s) - s^{n-1}f(0^+)$ $-\cdots - sf^{(n-2)}(0^+) - f^{(n-1)}(0^+)$
4. Integral, (7.31)	$\mathcal{L}\left[\int_0^t f(\tau)d\tau\right] = \frac{F(s)}{s}$
5. Real shifting, (7.22)	$\mathcal{L}[f(t-t_0)u(t-t_0)] = e^{-t_0s}F(s)$
6. Complex shifting, (7.20)	$\mathcal{L}[e^{-at}f(t)] = F(s+a)$
7. Initial value, (7.36)	$\lim_{t\to 0^+} f(t) = \lim_{s\to\infty} sF(s)$
8. Final value, (7.39)	$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$
9. Multiplication by t, (7.34)	$\mathcal{L}[tf(t)] = -\frac{dF(s)}{ds}$
10. Time transformation, (7.42) ($a > 0; b \geq 0$)	$\mathcal{L}[f(at-b)u(at-b)] = \frac{e^{-sb/a}}{a}F\left(\frac{s}{a}\right)$
11. Convolution	$\mathcal{L}^{-1}[F_1(s)F_2(s)] = \int_0^t f_1(t-\tau)f_2(\tau)\,d\tau$ $= \int_0^t f_1(\tau)f_2(t-\tau)d\tau$
12. Time periodicity $[f(t) = f(t+T), t \geq 0]$	$\mathcal{L}[f(t)] = \frac{1}{1-e^{-sT}}F_1(s)$, where $F_1(s) = \int_0^T f(t)e^{-st}\,dt$